

$$F(y) = \int_0^{\infty} \underbrace{\frac{\arctg(xy)}{x(1+x^2)}}_{f_y(x)} dx. \text{ za katere } y \text{ } \exists F?$$

Problem: 1) $x=0$ pol?
2) $x \rightarrow \infty$

$$1) \lim_{x \rightarrow 0} f_y(x) = \lim_{x \rightarrow 0} \frac{\arctg(xy)}{x(1+x^2)} = \lim_{x \rightarrow 0} \frac{\arctg(xy)}{x} \stackrel{L'H}{=} \lim_{x \rightarrow 0} \frac{y}{1+x^2 y^2} = y$$

$\rightarrow f_y$ ima zvezno razširitev v $x=0$, in sicer $f_y(0)=y$

$$2) f_y(x) = \frac{\arctg(xy)}{x(1+x^2)} < \frac{\frac{\pi}{2}}{x^3}$$

$$F(y) = \underbrace{\int_0^1 f_y(x) dx}_{\exists \in \mathbb{R}} + \int_1^{\infty} f_y(x) dx$$

$$\int_1^{\infty} f_y(x) dx < \frac{\pi}{2} \int_1^{\infty} x^{-3} dx < \infty$$

$\Rightarrow F(y) \exists$ za vse $y \in \mathbb{R}$.

$$F(y) = \frac{d}{dy} \int_0^{\infty} \frac{\arctg(xy)}{x(1+x^2)} dx = \int_0^{\infty} \frac{1}{x(1+x^2)} \cdot \frac{x}{1+x^2 y^2} dx = \int_0^{\infty} \frac{A}{1+x^2} dx + \int_0^{\infty} \frac{B}{1+x^2 y^2} dx =$$

$$\frac{A}{1+x^2} + \frac{B}{1+x^2 y^2} = \frac{1}{(1+x^2)(1+x^2 y^2)}$$

$$A + A x^2 y^2 + B + B x^2 = 1$$

$$A + B = 1$$

$$A y^2 + B = 0$$

$$A(1-y^2) = 1 \rightarrow A = \frac{1}{1-y^2}; y \neq \pm 1 \quad B = -\frac{y^2}{1-y^2}; y \neq \pm 1$$

$$= A \arctg x \Big|_0^{\infty} + \frac{1}{y} B \arctg xy \Big|_0^{\infty} = \frac{1}{1-y^2} \frac{\pi}{2} - \frac{y}{1-y^2} \frac{\pi}{2} = \frac{\pi}{2} \cdot \frac{1}{1+y}$$

F liha $\Rightarrow F'$ soda
če $y > 0$

$$F(y) = \frac{\pi}{2} \int \frac{1}{1+y} dy = \frac{\pi}{2} \ln|1+y| + c$$

$$F(0) = 0 = \frac{\pi}{2} \ln(1+0) + c \rightarrow \underline{c = 0}$$

Moramo še dokazati:

i) F zvezna

ii) da $\frac{d}{dy} \int f = \int \frac{\partial f}{\partial y}$

i) Preverjamo en. konv. F za $y \in [y_0+1, y_0+1]$; $a > 0$

$$\left| \int_c^\infty \frac{\arctan xy}{x(1+x^2)} dx \right| \leq \underbrace{\int_c^\infty \frac{\arctan(xy)}{xy} dx}_{\text{omej. s } C} \cdot \frac{y}{1+x^2} = Cy \int_c^\infty \frac{1}{1+x^2} dx$$

če je y dovolj blizu y_0 ,
je ocena majhna $\Rightarrow F$ enak. konv. na $y \in [a, \infty)$
 $\Rightarrow F$ je lokalno enakom. zvezna ✓

ii) preverjamo lok. en. zveznost $\int \frac{\partial f_y}{\partial y} dx$:

$$\int \frac{\partial f_y}{\partial y} dx = \int_0^\infty \underbrace{\frac{1}{(1+x^2)(1+x^2y^2)}}_{\wedge \frac{1}{1+x^2}} dx$$

$\frac{1}{1+x^2}$ integrab., neodv. od $y \Rightarrow \int_0^\infty \frac{\partial f_y}{\partial y} dx$ je enak. konv.

$$\Rightarrow F(y) = \frac{\pi}{2} \ln(1+y); y > 0 \text{ in } F(y) = -F(y)$$

$$\rightarrow F(y) = \text{sgn}(y) \frac{\pi}{2} \ln(1+|y|)$$

$$y(x) = \int_0^{\infty} \frac{e^{-xt}}{1+t^2} dt, \quad x > 0 \quad \text{izračunaj } y'' + y.$$

$$y'(x) \stackrel{1)}{=} - \int_0^{\infty} \frac{t e^{-xt}}{1+t^2} dt$$

$$y''(x) \stackrel{2)}{=} \int_0^{\infty} \frac{t^2 e^{-xt}}{1+t^2} dt$$

$$y + y'' = \int_0^{\infty} \frac{e^{-xt}(1+t^2)}{1+t^2} dt = \int_0^{\infty} e^{-xt} dt = + \frac{1}{x} \quad \uparrow \quad x \neq 0$$

$$I(a) = \int_{-\infty}^{\infty} \frac{dx}{(x^2+a)^2} \quad \text{Za katere } a \text{ } I \exists? \text{ Izračunaj}$$

$$1.) \text{ če } a > 0: \text{ ni polov } \Rightarrow \text{ konv., ker } \int_c^{\infty} \frac{1}{x^2} dx \exists$$

$$2.) I(0) = \int_{-\infty}^{\infty} \frac{dx}{x^2} \Rightarrow I \text{ ne } \exists, \text{ ker v ničli } x^4 \text{ prehitro raste}$$

$$3.) a < 0: I(-1) = \int_{-\infty}^{\infty} \frac{dx}{(x-1)(x+1)^2} \quad \text{Ni int. zaradi}$$

$$a > 0:$$

$$I(a) = \int_{-\infty}^{\infty} \frac{dx}{(x^2+a)^2}$$

$$J(a) = \int_{-\infty}^{\infty} \frac{dx}{x^2+a} = \int_{-\infty}^{\infty} \frac{dx}{a\left(\left(\frac{x}{\sqrt{a}}\right)^2+1\right)} = \int_{-\infty}^{\infty} \frac{\sqrt{a} du}{a(1+u^2)} = \frac{1}{\sqrt{a}} \operatorname{arctg} u \Big|_{-\infty}^{\infty} = \frac{\pi}{a}$$

$$u = \frac{x}{\sqrt{a}}$$

$$du = \frac{1}{\sqrt{a}} dx$$

$$J'(a) = -I(a)$$

$$\boxed{I(a) = + \frac{\pi}{2} a^{-3/2}}$$